

DAY 1 WEEK 1

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PROJECT: Battery SoC Estimation & Remaining Charging-Time Prediction Using Machine Learning

Day 1: Problem Understanding & System Definition

1. Problem Statement Understanding

- Accurate Battery State of Charge (SoC) estimation and remaining charging-time prediction are important for EV battery charging systems. The problem is to use measured battery data such as voltage, current, power, temperature, and charging history to build ML regression models that predict these two values.
- Battery SoC and remaining charging time cannot be directly measured by a single sensor during charging.
- Both outputs must be inferred from measurable charging signals and historical information.
- The project is framed as a supervised regression problem because both targets are continuous.
- The objective is to learn relationships between charging measurements and the two target outputs.

2. Pipeline Diagram

Signal → Model → Prediction

Input Signals: Battery Voltage, Charging Current, Charging Power (Derived), Battery Temperature, Charging History, Time / Time Stamp

Machine Learning Model (Regression Model) — Linear Regression / Decision Tree / Random Forest / XGBoost (Optional: ANN)

Prediction: Battery SoC (%) and Remaining Charging Time (estimated) — two regression heads, one shared feature set.

End-to-End Workflow:

1. Problem Framing – Define SoC and charging-time prediction as an ML regression task.

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2. Data Understanding – Audit and understand the provided battery charging dataset.

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3. Data Pipeline – Build a reproducible data-cleaning and EDA pipeline.

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4. Model Training & Comparison – Train and compare baseline, RF, XGBoost and optional ANN models.

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5. Model Evaluation – Evaluate models using MAE, RMSE, and R^2 for both target variables separately.

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6. Error Analysis – Analyse prediction errors by charging phase and SoC range, not just overall accuracy.

3. Questions for Mentor

1. What are the main columns in the dataset?
2. What are the units of each column?
3. How is SoC calculated?
4. How is charging time defined?
5. How should we split the data for training and testing?